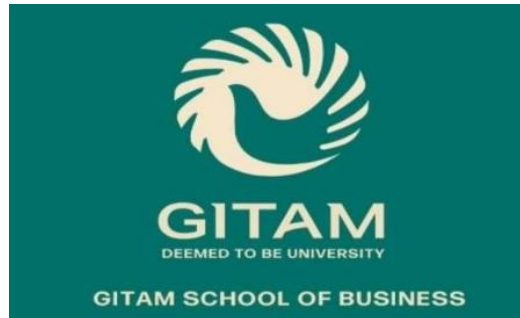


USER BEHAVIOR ANALYSIS

Team - 3



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1. Introduction

User Behavior Analysis studies how users interact with a website by analyzing metrics such as pages viewed, time spent on site, scroll depth, and conversion activity. It helps organizations understand user engagement, identify problem areas, and improve website performance.

In this study, R programming is used to analyze user behavior data through descriptive statistics and visualizations. The analysis aims to evaluate engagement patterns and understand how user actions influence conversions.

2. Literature Review

Digital analytics focuses on measuring user engagement and behavior to improve website performance. Metrics such as session duration, bounce rate, and conversion rate are commonly used to evaluate platform effectiveness.

Key Metrics Used:

- Pages Viewed
- Time on Site
- Scroll Depth
- Landing Page
- Exit Page
- Conversion Status

These metrics are widely adopted in web analytics tools such as Google Analytics and Adobe Analytics.

3. Data Collection

3.1 Data Source

The data set used in this study consists of user behavioral data set collected from a source. Each record represents one session data.

3.2 Data Loading

```
behavior <- read.csv("user_behavior_sample.csv")
```

4. Methodology

This study follows a systematic R-based analytics workflow to analyze user behavior data. The methodology includes data preparation, descriptive analysis, visualization, and correlation analysis to understand engagement and conversion behavior.

4.1 Analytical Framework

The user behavior analysis is conducted using the following steps: Data loading and validation Computation of engagement metrics Page-level performance analysis Visualization of user behavior Correlation analysis between engagement variables R is used as the primary analytical tool due to its strong statistical and visualization capabilities.

4.2 Data Preparation

Before analysis, the dataset is checked for structure and missing values. Numeric variables are prepared for statistical computation, and categorical variables are used for grouping analysis. View dataset structure

```
str(behavior)
##      'data.frame':      1000 obs. of      9 variables:
##  $ user_id      : chr   "user_1" "user_2" "user_3" "user_4" ...
##  $ session_id   : chr   "session_1415" "session_1463" "session_1179"
##  "session_1526"
##  $ landing_page : chr    "/home"  "/home"  "/home"  "/home"  ...
##  $ exit_page    : chr    "/home"  "/blog"  "/cart"  "/blog"  ...
##  $ pages_viewed : int      4  7  7  7  4  10  2  9  3  8  ...
##  $ time_on_site : int   916 1448 112 493 382 1475 1522 1504 509 65  ...
##  $ scroll_depth  : num     0.839 0.869 0.113 0.134 0.369  ...
##  $ conversion   : int      0  0  0  0  0  0  0  0  1  0  ...
##  $ page_sequence: chr              "/checkout|/checkout|/product-
##  details|/products|/cart|/contact|/blog|/about"              "/cart|/faq"
##  "/contact|/faq|/about|/contact|/about|/blog|/blog"              "/product-
##  details|/about|/faq|/product-details|/checkout|/blog" ...
```

Check missing values

```
colSums(is.na(behavior))
##      user_id      session_id  landing_page      exit_page  pages_viewed
##           0           0           0           0           0
##  time_on_site  scroll_depth      conversion  page_sequence
##           0           0           0           0
```

4.3 Engagement Metrics Calculation

User engagement is measured using average values of key behavioral variables. These metrics provide a high-level summary of overall user interaction with the platform.

Metrics Used: Average Pages Viewed

Average Time on Site

Average Scroll Depth

Conversion Rate

```
engagement_summary <- behavior %>%
  summarise(
    avg_pages_viewed = mean(pages_viewed, na.rm = TRUE),
    avg_time_on_site = mean(time_on_site, na.rm = TRUE),
    avg_scroll_depth = mean(scroll_depth, na.rm = TRUE),
    conversion_rate = mean(conversion, na.rm = TRUE))
engagement_summary
##      avg_pages_viewed avg_time_on_site avg_scroll_depth conversion_rate
## 1           5.987      902.022      0.5513711          0.146
```

5. Analysis and Results

5.1 Page-Level Performance Analysis

Landing page and exit page analyses are performed to evaluate user navigation patterns. This helps identify entry points with high traffic and exit points with potential usability issues.

Landing Page Analysis

```
landing_page_analysis <- behavior %>%
  group_by(landing_page) %>%
  summarise(
    sessions = n(),
    avg_time = mean(time_on_site, na.rm = TRUE),
    conversion_rate = mean(conversion, na.rm = TRUE))
landing_page_analysis
##      #      A      tibble: 4 × 4
##      landing_page      sessions      avg_time      conversion_rate
##      <chr>      <int>      <dbl>      <dbl>
## 1 /about      103      840.      0.0777
## 2 /blog       199      913.      0.136
## 3 /home       387      907.      0.163
## 4 /products   311      910.      0.154
```

Exit page analysis

```
exit_analysis <- behavior %>%  
  group_by(exit_page) %>%  
  summarise( exits = n(),  
    exit_rate = n() / nrow(behavior) ) %>% arrange(desc(exits))
```

5.2 Descriptive Analysis

```
behavior <- read.csv("user_behavior_sample.csv")  
  
summary(behavior[, c("pages_viewed", "time_on_site", "scroll_depth")])  
  
##   pages_viewed   time_on_site   scroll_depth  
##   Min.      : 1.000   Min.      : 33.0   Min.      :0.1015  
##   1st Qu.: 4.000   1st Qu.: 481.0   1st Qu.:0.3429  
##   Median : 6.000   Median : 903.5   Median :0.5377  
##   Mean    : 5.987   Mean    : 902.0   Mean    :0.5514  
##   3rd Qu.: 7.000   3rd Qu.:1345.2   3rd Qu.:0.7680  
##   Max.    :17.000   Max.     :1798.0   Max.     :0.9997
```

Interpretation

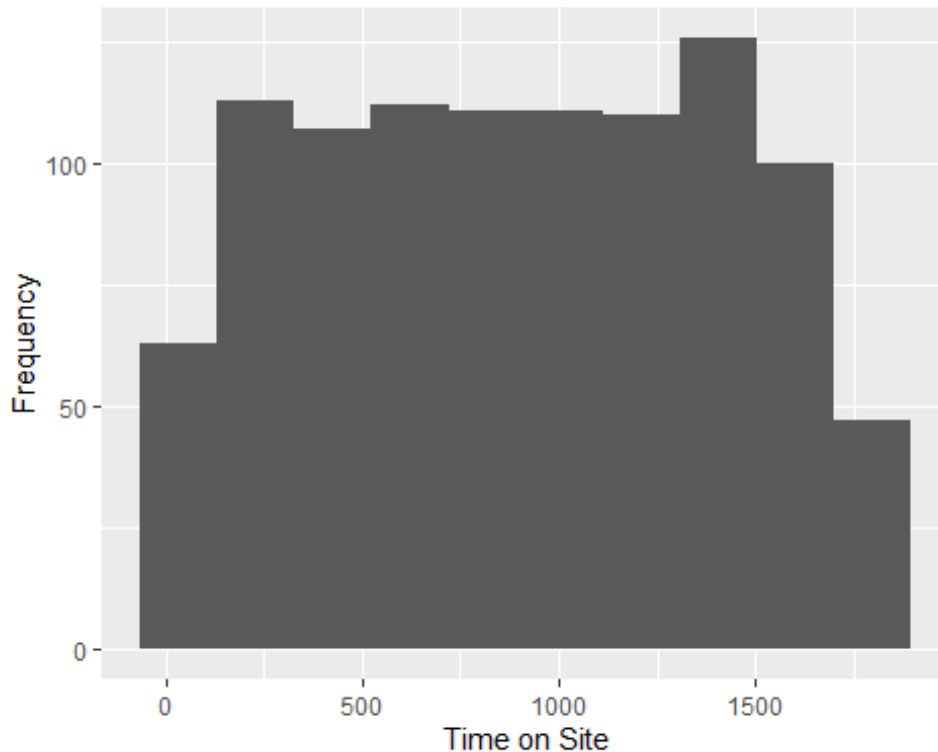
The descriptive statistics show that users view an average of about 6 pages and spend approximately 15 minutes on the website, indicating moderate to good engagement. The average scroll depth is around 55%, suggesting that users read more than half of the content on a page. Overall, the data reflects healthy user interaction with the website.

5.3 Visualization Techniques

Visualization techniques are used to understand data distribution and relationships between variables.

5.3.1 Histogram – Time on Site

```
ggplot(behavior, aes(x = time_on_site))  
geom_histogram(bins = 10)  
labs(x = "Time on Site", y = "Frequency")
```

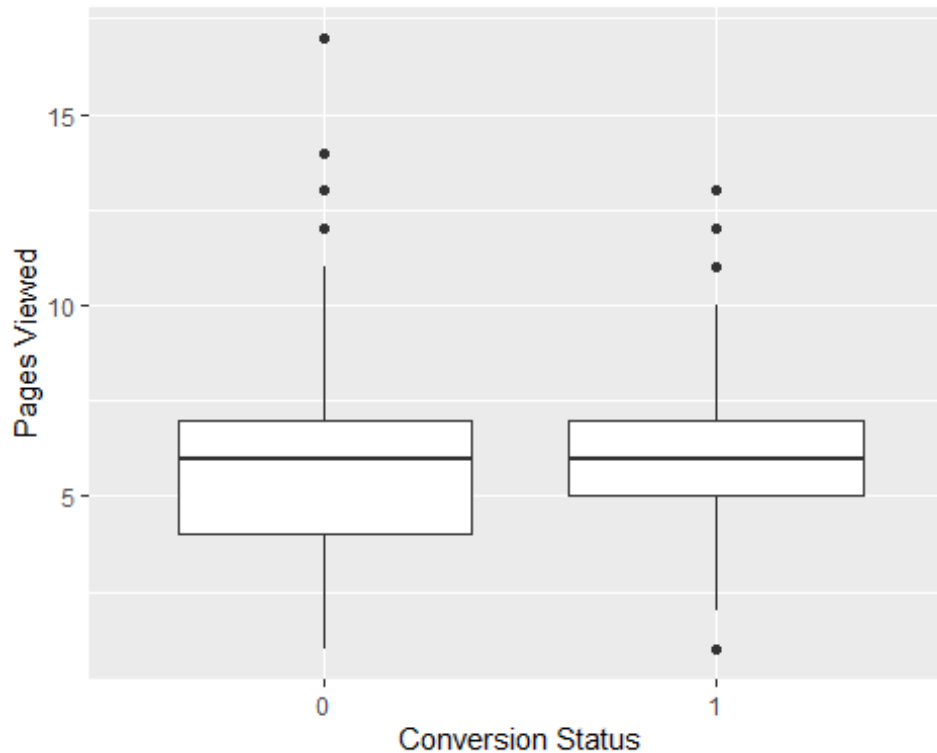


Interpretation

The histogram illustrates the distribution of users based on the amount of time spent on the website during a session. The data shows that most users spend a moderate amount of time on the site, while fewer users remain for very long durations. This uneven distribution indicates variability in user engagement levels, where some users browse quickly and exit, while others stay longer to explore content in detail. The visualization suggests that time spent on the site alone does not guarantee deeper engagement or conversion, as users display diverse browsing behaviors.

5.3.2 Box Plot – Pages Viewed by Conversion

```
ggplot(behavior, aes(x = as.factor(conversion), y = pages_viewed))  
  geom_boxplot()  
  labs(x = "Conversion Status", y = "Pages Viewed")
```

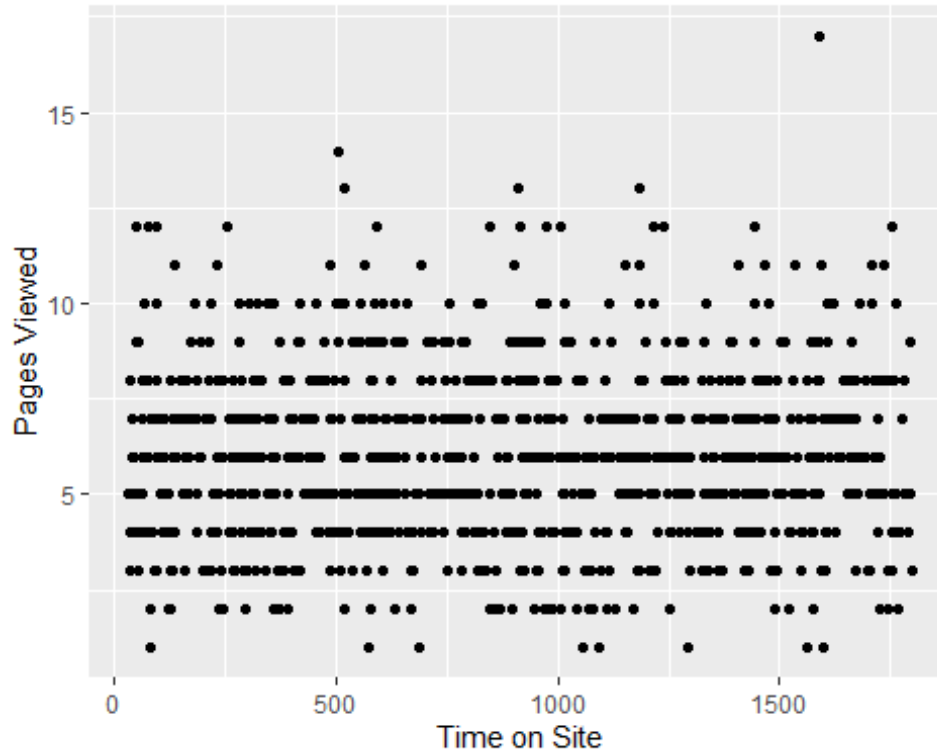


Interpretation

The box plot illustrates a significant overlap in behavior, with converting users viewing an average of 6.25 pages compared to 5.94 pages for non-converters. This slight difference indicates that the volume of pages viewed is not a definitive predictor of a successful transaction. The data suggests that high engagement is common across all users, meaning conversion is likely driven by the content relevance of specific pages rather than the total quantity consumed.

5.3.3 Scatter Plot – Engagement Relationship

```
ggplot(behavior, aes(x = time_on_site, y = pages_viewed))  
geom_point()  
labs(x = "Time on Site", y = "Pages Viewed")
```



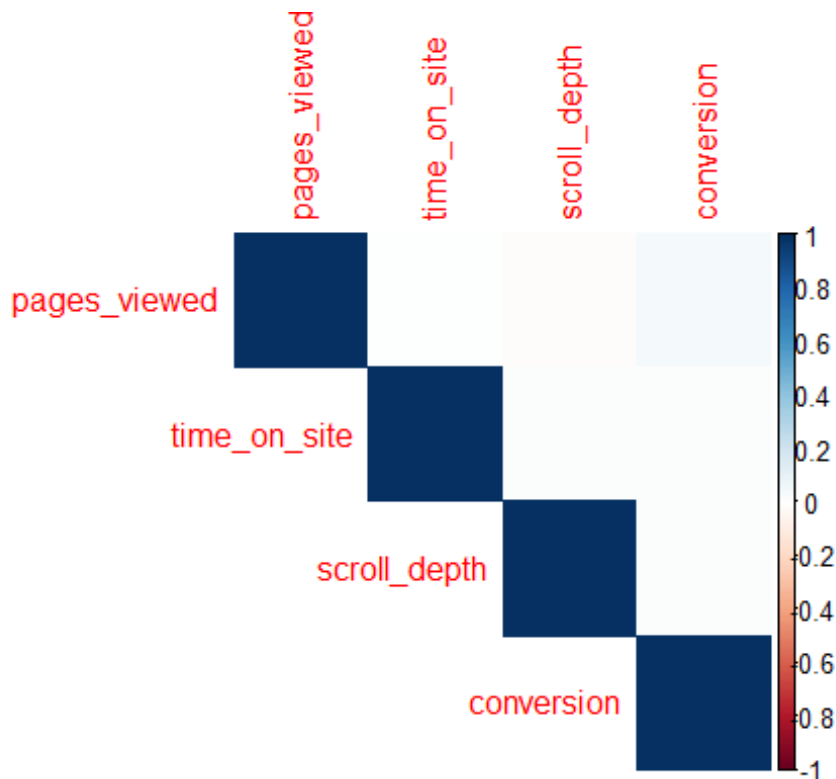
Interpretation

The scatter plot displays a widely dispersed cloud of points with no discernible pattern, confirming a near-zero correlation between the time spent on the site and the number of pages viewed. This independence indicates that users browse at vastly different speeds—some consume many pages quickly, while others linger for long periods on just a few. Consequently, “Time on Site” is not a reliable predictor of how much content a user has actually seen.

5.4 Correlation Analysis

Correlation analysis is used to measure the strength and direction of relationships between engagement metrics and conversion behavior.

```
correlation_matrix <- cor(behavior[, c("pages_viewed", "time_on_site", "scroll_depth", "conversion")],  
use = "complete.obs")  
corrplot(correlation_matrix, method = "color", type = "upper")
```



Interpretation

The correlation analysis demonstrates a negligible relationship between user engagement metrics and conversion outcomes, with all correlation coefficients near zero. This statistical independence indicates that neither the duration of a visit nor the depth of scrolling serves as a reliable predictor for purchasing behavior. Consequently, conversions appear to be driven by factors external to these specific engagement metrics, suggesting that “more” activity does not necessarily lead to “better” outcomes.

5.5 Multiple Regression Analysis

Multiple regression is a statistical method that predicts a single dependent variable using two or more independent variables.

```
multi_model <- lm(pages_viewed ~ time_on_site + scroll_depth, data =
behavior)
summary(multi_model)

##
## Call:
## lm(formula = pages_viewed ~ time_on_site + scroll_depth, data = behavior)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.0420 -1.9382 -0.0115  1.0570 11.0091
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  6.059e+00  2.123e-01  28.533  <2e-16 ***
## time_on_site  4.066e-06  1.424e-04   0.029   0.977
## scroll_depth -1.372e-01  2.802e-01  -0.490   0.624
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.266 on 997 degrees of freedom
## Multiple R-squared:  0.0002412, Adjusted R-squared:  -0.001764
## F-statistic: 0.1202 on 2 and 997 DF,  p-value: 0.8867
```

Interpretation

This regression model was used to understand whether time on site and scroll depth together influence the number of pages viewed by users. The results show that neither time on site nor scroll depth has a significant impact on pages viewed, as both variables have very high p-values (greater than 0.05). The R-squared value is almost zero, which means the model explains almost none of the variation in pages viewed. The overall F-test is also not significant, indicating that the model does not fit the data well.

5.6 Chi-Square Test

Chi-square is a statistical measurement that shows how well a model fits the observed data.

```
chisq.test(table(behavior$exit_page, behavior$conversion))  
  
##  
## Pearson's Chi-squared test  
##  
## data:  table(behavior$exit_page, behavior$conversion)  
## X-squared = 0.30913, df = 4, p-value = 0.9892
```

Interpretation

The chi-square test between exit page and conversion resulted in a p-value of 0.9892, which is greater than 0.05. This indicates that there is no significant association between the exit page and user conversion. In simple terms, the page from which users exit the website does not influence whether they convert or not.

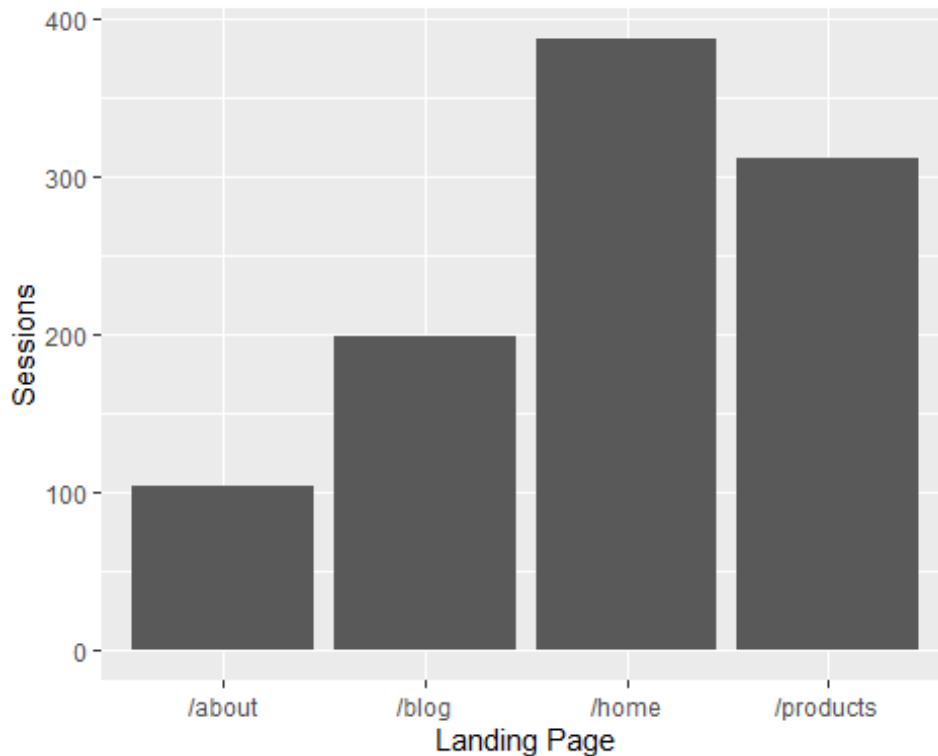
5.7 Tools and Packages Used

R – Primary analytical tool dplyr – Data manipulation ggplot2 – Data visualization corrplot – Correlation visualization

The methodology combines statistical analysis and visualization techniques to examine user behavior patterns. By leveraging R-based tools, the study provides a structured and data-driven approach to understanding engagement and conversion behavior.

5.8 Landing Page Analysis

```
landing_page_analysis <- behavior %>%  
  group_by(landing_page) %>%  
  summarise(sessions = n())  
  
ggplot(landing_page_analysis, aes(x = landing_page, y = sessions)) +  
  geom_bar(stat = "identity") +  
  labs(x = "Landing Page", y = "Sessions")
```

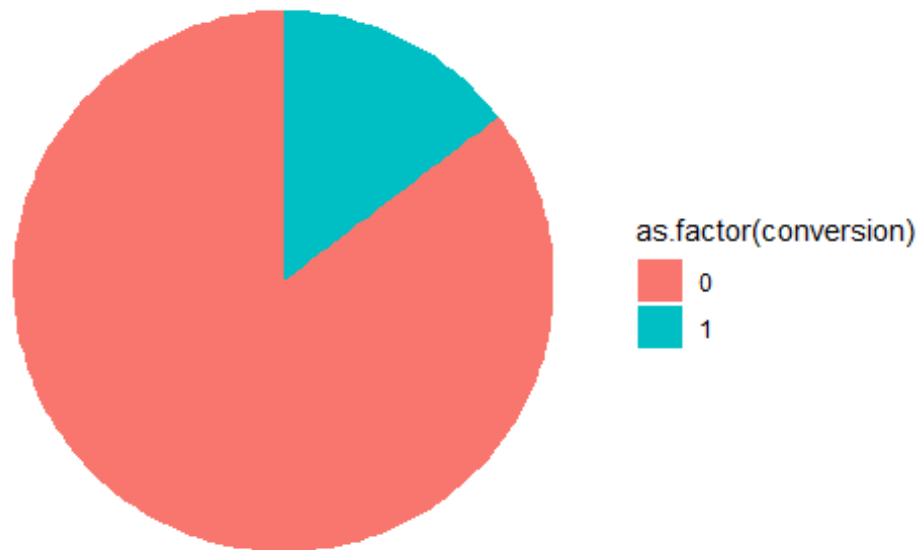


Interpretation

The landing page analysis highlights how users enter the website. The Home page records the highest number of sessions, making it the primary entry point for users. This is followed by the Products and Blog pages, indicating that users also arrive through product exploration and informational content. The About page has the lowest number of sessions, suggesting it is less commonly used as an entry point. Overall, the analysis shows that pages with higher session counts play a crucial role in shaping first impressions and user journeys, and optimizing these pages is essential for improving engagement and conversion outcomes.

5.9 Conversion Distribution

```
conversion_data <- behavior %>%  
  count(conversion)  
ggplot(conversion_data, aes(x = "", y = n, fill = as.factor(conversion))) +  
  geom_bar(stat = "identity", width = 1) +  
  coord_polar("y") +  
  theme_void()
```



Interpretation

The pie chart reveals that the vast majority of visitors, 85.4% (854 users), left the site without converting, while only 14.6% (146 users) completed a desired action. This significant imbalance highlights that while the site successfully attracts traffic, most journeys end without a transaction. Consequently, the primary optimization opportunity lies in improving the user funnel to capture the large pool of engaged but non-converting visitors.

5.10 Overall Interpretation

The analysis shows that higher engagement metrics such as time on site and pages viewed are positively associated with conversion. Landing pages with higher session counts indicate strong user interest, while exit page analysis highlights potential drop-off points. Correlation results suggest that engagement variables are important drivers of conversion behavior.

6. Conclusion and Recommendations

6.1 Conclusion

Conclusion This study demonstrates the effectiveness of R-based user behavior analysis in identifying engagement patterns and conversion drivers.

6.2 Recommendations

Optimize high-exit pages to reduce user drop-off

Improve content on high-performing landing pages

Focus on increasing session duration to improve conversion rates.

7. References

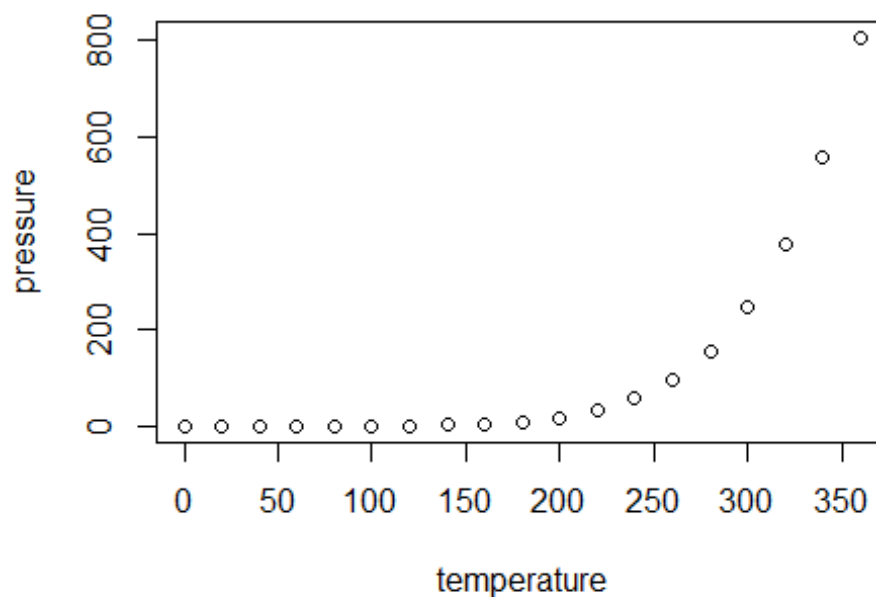
Kaushik, A. (2010). Web Analytics 2.0. Wiley

Wickham, H. (2016). ggplot2: Elegant Graphics for Data Analysis. Springer

Google Analytics Documentation (2024)

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.